

Can It Ford? Query-Conditioned, Physically Viable World Models for Autonomous-Vehicle Flood Traversability from Gaussian Splatting

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Abstract—Flooded roads are a leading cause of vehicle fatalities each year, killing drivers who cannot judge how deep the water is, how fast it moves, or what the road beneath it is doing. Neither human drivers nor automated vehicles can read these things from appearance alone. World models that predict only appearance inherit the same limitation, since they can render an outcome that looks plausible but is physically wrong. This motivates physically viable world models, which preserve the latent physics that actually governs what happens when a vehicle enters floodwater. This project asks a deliberately narrow question: can an autonomous vehicle ford a specific flooded road? It then identifies the simplest physical abstraction sufficient to answer that query reliably, rather than the most detailed simulation that can be built. We reconstruct a flooded-road scene using 3D Gaussian splatting (gsplat). Following PhysGaussian, the Gaussian kernels seed a continuum representation that initializes a Material Point Method (MPM) simulation in the Genesis physics engine, where rigid-MPM coupling allows a rigid vehicle to interact with weakly compressible floodwater. We compare three levels of physical fidelity: a static depth threshold, a depth-velocity stability criterion, and, where it can be stabilized within the project timeline, a full coupled MPM simulation. Where feasible, we expect the coupled model to reproduce empirical instability thresholds, with buoyancy near published float depths and sliding governed by the depth-velocity product. Our central claim is that the minimal sufficient abstraction is query-dependent: a simple threshold suffices for deep, still water, while fast, shallow flow requires the coupled model to return the correct answer. This work delivers a reconstruct-to-decide pipeline and a concrete instantiation of query-conditioned, physically viable world models, and a small, testable step toward autonomous systems that can physically reason about a hazard before committing to it.

Index Terms—physically viable world models, Gaussian splatting, Material Point Method, flood traversability, autonomous vehicles, abstraction selection

I. INTRODUCTION

Flooded roads kill more people in the United States each year than any other flood-related hazard, and most of those

This work was supported by the National Science Foundation under NSF REU Site: Cyberinfrastructure Research for Societal Advancement (SCIFE-AI), Award #2447887.

deaths involve a vehicle driven into water whose depth, flow velocity, or submerged road condition could not be judged from the surface [1]. This is not only a human-driver problem. An autonomous vehicle facing the same scene has the same information gap, and a world model that predicts only visual appearance can render a plausible-looking outcome that is physically wrong: it may show a vehicle continuing forward with no indication that the same maneuver would, in reality, cause loss of traction or flotation.

Physically viable world models (PVWM) address this by preserving the latent physics that governs an action’s outcome rather than only its appearance [2]. The PVWM framework identifies automatic abstraction selection, choosing how much physical fidelity a given query actually requires, as a central open problem, and leaves the orchestrator that would make that choice unimplemented [2].

This project instantiates that orchestrator for one deliberately narrow query: *can this specific autonomous vehicle ford this specific flooded road right now?* Rather than building the highest-fidelity simulation possible, the goal is to identify the simplest physical abstraction sufficient to answer that query correctly, and to characterize where a cheaper abstraction silently gives the wrong answer. Figure 1 summarizes the full reconstruct-to-decide pipeline this paper describes.

The remainder of this paper is organized as follows. Section II reviews prior flood-vehicle stability work and the reconstruction and simulation methods this project builds on. Section III describes the pipeline and the three-level abstraction ladder in detail. Section IV presents current results. Section V concludes. Section VI outlines future work.

II. PRIOR WORK

A. Flood-Vehicle Stability Criteria

Two families of empirical criteria describe when floodwater destabilizes a vehicle. The Australian Rainfall and Runoff (AR&R) Project 10 guidelines define a depth-velocity hazard scalar, $H = D \times V$ (m²/s), with published class-specific

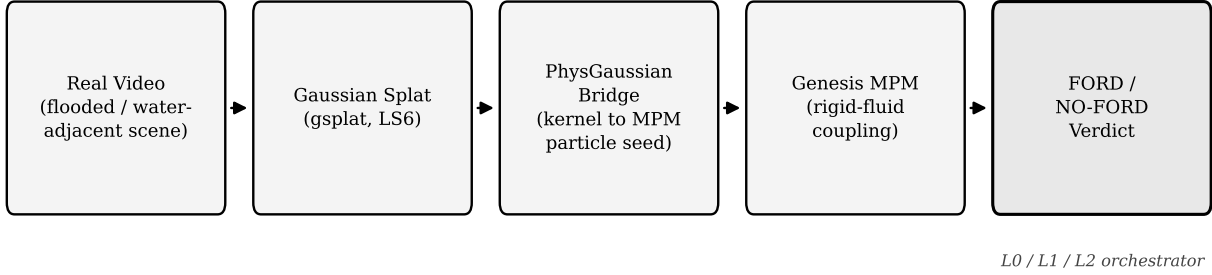


Fig. 1. The reconstruct-to-decide pipeline: a real video is reconstructed as a Gaussian splat, bridged to Material Point Method particles following PhysGaussian, simulated in Genesis with rigid-fluid coupling, and resolved to a FORD/NO-FORD verdict by the L0/L1/L2 abstraction orchestrator.

thresholds [3]. Smith, Modra, and Felder report full-scale flotation and sliding stability curves derived from physical vehicle testing [4]. Both are scalar, single-number criteria: neither represents the direction of applied force, and neither was derived to predict the persistent lateral drift this project observes at some phase-space points (Section IV).

B. Physics-Grounded Scene Reconstruction

3D Gaussian Splatting (3DGS) reconstructs a photorealistic, explicit scene representation from ordinary multi-view video [5]. PhysGaussian extends this representation with physical simulation by treating each Gaussian kernel’s covariance as a seed for a continuum Material Point Method (MPM) particle volume, allowing a trained splat to drive a physics simulation directly rather than only a renderer [6]. The simulation itself runs in Genesis, an open-source, solver-agnostic physics engine [7]. [FLAG: No publicly available, drop-in flood-video-to-Genesis-MPM pipeline exists as of this writing; PhysGaussian’s own reference implementation targets a Warp MPM solver, not Genesis. Confirm this framing is still accurate before submission.]

C. Why a Single Scalar Threshold Is Insufficient

Both L0 and L1 are deterministic functions of published thresholds, not simulation output: L0 applies a single fixed depth cutoff [1], and L1 applies the AR&R two-part depth-and-hazard criterion per vehicle class [3]. Evaluating both across the same depth-velocity grid, Fig. 2 shows that L0 alone is markedly over-conservative, and that AR&R’s own published per-class caps (Fig. 3) already diverge from one another well before any coupled simulation enters the picture. Neither figure required simulated data to produce; both are direct evaluations of cited formulas and are included here to motivate why a coupled physical model is worth the added cost.

D. Forward and Inverse Property Estimation

Hsiao and Kumar demonstrate the inverse direction of this same reconstruction-to-physics pipeline: given only visual observation, a NeRF-based reconstruction paired with Bayesian

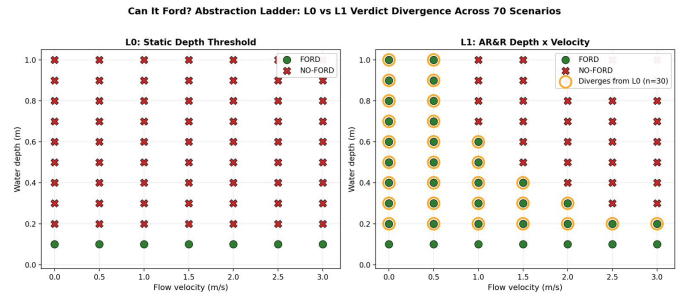


Fig. 2. L0 (fixed depth threshold) versus L1 (AR&R depth-velocity hazard scalar) across 70 (depth, velocity) scenarios. Both panels are direct evaluations of published formulas [1], [3], not simulation output. Orange circles mark the 30 scenarios where L1 permits crossing but L0 alone would not.

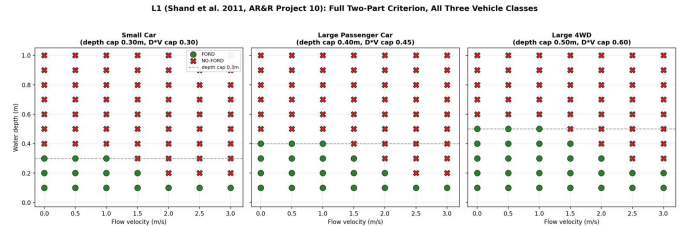


Fig. 3. The L1 criterion evaluated per AR&R’s own three published vehicle classes [3]. [FLAG: AR&R’s class labels (Small Car / Large Passenger Car / Large 4WD) do not map one-to-one onto the sedan/SUV/pickup NHTSA/SAE classes used elsewhere in this paper (Table II); state the mapping explicitly in the caption or body text before submission, or a reader will reasonably assume they are the same three classes.]

optimization recovers a granular material’s friction angle to within roughly two degrees of ground truth [8]. This project addresses the complementary forward direction: given a known scene and known flood hydraulic conditions, what is the traversability verdict under full fluid-rigid-body dynamics? Together, the two directions describe a closed perception-to-action loop, one of the auditability properties PVWM requires [2].

TABLE I
ABSTRACTION LADDER DEFINITIONS

Level	Basis	Definition
L0	Depth	Fixed depth cutoff (≈ 0.15 m [1]), independent of velocity or vehicle class.
L1	$D \times V$	$H = D \times V$ vs. class threshold ($0.60 \text{ m}^2/\text{s}$, 4WD default) [3]. V is flow velocity, not vehicle speed; mass and force direction are not represented.
L2	MPM	Genesis rigid-MPM simulation; NO-FORD when lateral drift exceeds a threshold. [FLAG: Threshold currently 0.05 m with no independent citation; being reframed relative to Smith-Modra-Felder (2019) Eq. 6.]

III. APPROACH

A. Pipeline

The reconstruct-to-decide pipeline (Fig. 1) has four stages: (1) a real flooded or water-adjacent road scene is captured on video and reconstructed in 3D via Gaussian splatting on an NVIDIA A100 GPU (TACC Lonestar6); (2) following the PhysGaussian extraction pattern, trained Gaussian kernels are opacity-filtered and converted into an initial MPM particle volume; (3) the particle volume, together with a rigid-body vehicle proxy, is simulated in the Genesis physics engine using weakly-compressible MPM water and rigid-MPM coupling; (4) the simulation’s displacement output is passed through a query-conditioned abstraction-selection orchestrator, `orchestrate_ford_query()`, which returns a FORD/NO-FORD verdict at the cheapest level of fidelity sufficient to answer correctly.

B. Three-Level Abstraction Ladder

Three levels of physical fidelity are compared directly against one another on the same phase-space grid of (depth, velocity) conditions, summarized in Table I.

This instantiates the orchestrator PVWM leaves unimplemented: the empirical L1/L2 divergence zone characterized in Section IV defines the escalation boundary at which L1 alone is not sufficient and the query must be answered at L2.

A first-principles force balance motivates why L1’s scalar hazard measure is expected to be insufficient in the first place. Fig. 4 plots lateral hydrodynamic drag force against maximum available static friction at several water depths (left), and the resulting critical velocity above which drag is predicted to exceed friction, as a function of depth (right). [FLAG: Confirm the drag coefficient, frontal area, and friction coefficient used to generate this figure are the same cited values used elsewhere in the paper (Azhar et al. 2023 range for μ) before submission, and cite the drag-force equation this figure is built from explicitly in the caption or body text.]

C. Vehicle and Scene Representation

The vehicle is represented as a rigid body solidified from a surface Gaussian splat (`truck_trimmed.ply`, 191,107 points, no interior), since no watertight simulation-ready mesh currently exists in the pipeline. Three vehicle classes (sedan,

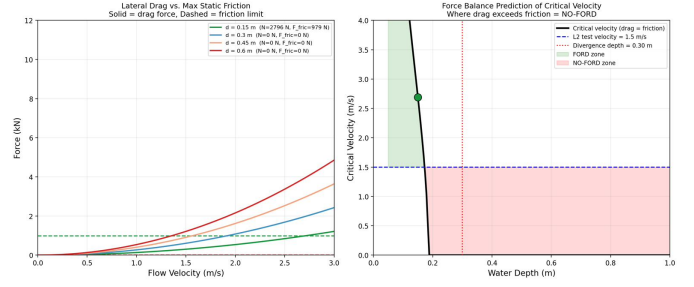


Fig. 4. Analytical force-balance calculation, not simulation output. Left: lateral drag force versus flow velocity at four water depths, against the maximum static friction limit (dashed). Right: the resulting critical velocity, above which drag exceeds friction, as a function of depth. [PLACEHOLDER: cite the specific drag/friction equation and coefficient values used before submission.]

SUV, pickup) are each a single rescaled base silhouette stretched to that class’s real bounding-box dimensions from NHTSA/SAE reference data [9]. This is stated explicitly as a limitation, not glossed over: all three classes share one silhouette, so cross-class comparisons in Section IV reflect scaled mass and footprint, not distinct vehicle geometry.

Solidifying a surface point cloud into a filled rigid body is grid-resolution-dependent in a way that is easy to miss. At MPM grid resolution $n_{\text{grid}}=64$, the column-fill solidification step correctly produces a solid body and the three densities in Table II are recovered. At $n_{\text{grid}}=128$, the same splat produces a hollow shell instead: verified directly, particle count scaled $4.31\times$ between the two resolutions rather than the $8\times$ a solid body requires, and 0 of 60 test runs at $n_{\text{grid}}=128$ passed a density plausibility check (all three classes’ apparent density roughly doubled, $1.84\text{--}1.87\times$, a resolution artifact, not a physical effect). Water resolution and vehicle solidity are governed by different length scales for this asset, raising grid resolution helps the former and silently breaks the latter. **Table II’s densities hold only at $n_{\text{grid}}=64$** and must not be read across to any higher-resolution run without re-solidifying the vehicle by a decoupled method first (canditates: fix the solidification pitch independent of grid pitch, densify the point cloud before solidifying, or move to a true watertight mesh, e.g., the NHTSA/NCAC crash-validated models in Section VI). This also settles, rather than merely defers, the question of testing 0.05 m depth: at $n_{\text{grid}}=64$ it seeds zero water particle layers for the pickup class, and at $n_{\text{grid}}=128$ the vehicle itself is invalid, so 0.05 m is not reachable with the current vehicle representation at any grid setting tested.

IV. RESULTS

A. Scene Reconstruction Status

Structure-from-motion reconstruction (COLMAP) is complete for one real scene (“Drain A”), confirmed via a populated sparse point cloud and camera pose set on TACC Vista scratch storage. [FLAG: Gaussian-splat checkpoint training from this reconstruction, which runs separately on LS6, has not been

TABLE II
VEHICLE CLASS DENSITY AT $n_{\text{GRID}}=64$ (ONLY VALID GRID SETTING)

Class	Density (kg/m ³)	In heuristic band?
Sedan	293.55	Yes
Pickup	242.12	Yes
SUV	308.13	No (2.7% over, left as-is)

Verified against
docs/track1_v3_sweep_invalid_hollow_vehicle.md. Values
are not meaningful at $n_{\text{grid}}=128$ (see body text).

confirmed complete as of this draft. Do not describe Drain A as a fully trained, simulation-ready splat until that is verified live. A second scene (“Drain B”) has not yet been captured or reconstructed.]

B. Synthetic-Geometry Pilot Study

An early parametric comparison of L1 against a Genesis SPH solver (data/l2_results_from_wandb.csv) covers 9 unique (depth, velocity) conditions. L1 and L2 agreed at 5 of 9 (55.6%). The 4 divergence points, where L1 returned FORD but L2 returned NO-FORD, were (0.30 m, 1.5 m/s), (0.15 m, 1.5 m/s), (0.30 m, 1.0 m/s), and (0.30 m, 2.0 m/s). [FLAG: A previous draft of this subsection cited 23 pairs, 16 divergence points, and 30.4% agreement. Those specific numbers do not match this file (9 pairs, 4 divergences, 55.6% agreement) and were never traced to a source file before being written; treat them as unverified and superseded by the numbers above, which are read directly from the CSV. A separate file, data/phase_space_results.csv, has 23 unique (depth, velocity) pairs matching the old claim’s pair count, but carries only a single verdict column with no corresponding L1 value, so an agreement rate cannot be computed from it, and 15 of its 31 rows share a condition with another row under a different verdict (e.g., three separate (0.30 m, 1.5 m/s) runs returned FORD, NO-FORD, and NO-FORD with three different displacements), so it is not run-labeled cleanly enough to use as-is either. Before final submission: either reconcile which file (and which L1 calculation) produced the original 23/16/30.4% claim, or report only the verified 9-pair result above.]

A separate 4-point friction sweep at the (0.30 m, 1.5 m/s) divergence point (data/mu_sweep_results.csv) shows lateral displacement of 0.399, 0.3957, and 0.3953 m at coupling friction $\mu = 0.3, 0.5, 0.7$ respectively, essentially invariant across that range. At $\mu = 0.0$, displacement drops to 0.3283 m, a real difference, not noise. [FLAG: A previous draft described this as “friction-invariant across $\mu = 0.0$ to 0.7.” That overstates it: invariance holds from $\mu = 0.3$ to 0.7; $\mu = 0.0$ is measurably lower. The qualitative point, that a scalar hazard measure cannot represent this directional force, still stands, but state the range precisely. This is also only 4 points at a single condition; do not generalize beyond it without more data.]

Pending a fully reconciled dataset, Fig. 5 remains an explicitly labeled schematic, not a plotted result, showing

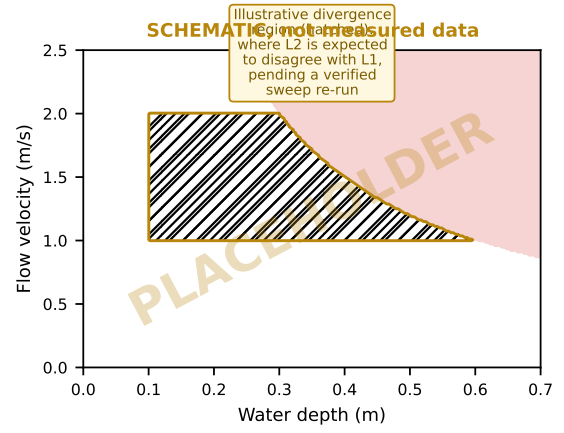


Fig. 5. Schematic illustration only, not measured or simulated data. Shows the conceptual region where the L1 hazard scalar is expected to disagree with a coupled L2 simulation; the verified 9-point pilot (Section IVA) supports the same qualitative shape.

only the conceptual shape of the claim. Do not remove the “SCHEMATIC” watermark until it is replaced by a figure built from the verified 9-point dataset above or a larger confirmed re-run.

C. Real-Simulation Sweep

A second, larger sweep (data/track1_sweep_v2/, files timestamped July 16, 2026 on Track 1) using the density-corrected vehicle at $n_{\text{grid}}=64$ exists on disk with a per-run manifest and per-run timeseries files, indexed by vehicle class, depth, and velocity. [FLAG: This sweep’s manifest has not been read in this session; its verdicts, its agreement with the pilot above, and whether it is the same sweep previously referred to in project notes as the “v2”/36-run sweep are all unconfirmed. A separate, later attempt at this sweep using $n_{\text{grid}}=128$ is confirmed invalid: the vehicle solidifies as a hollow shell at that resolution (Section IIIC), and its output must never be cited as a result. Read track1_sweep_v2/manifest.csv before writing this subsection for real.]

V. CONCLUSIONS

The synthetic-geometry pilot demonstrates, as a proof of concept, that a scalar depth-velocity hazard measure can disagree substantially with a coupled physical simulation, and that the disagreement is structural rather than a calibration artifact: no choice of coupling friction closes the gap at the divergence points identified in Section IV. This supports treating abstraction selection as query-dependent rather than assuming any single fixed-fidelity model is correct everywhere in the operating envelope. [FLAG: This is the pilot-study conclusion. Do not upgrade this to a claim about the real reconstructed scene until the real-scene sweep in Section IV is verified and the vehicle mesh/density limitation is either resolved further or explicitly scoped as a stated limitation.]

VI. FUTURE WORK

Several concrete extensions were identified during this project and are listed here roughly in order of expected impact relative to effort.

- **Crash-validated vehicle geometry.** NHTSA/NCAC publishes free, government-funded, crash-validated finite-element vehicle models (e.g., a 2010 Toyota Yaris sedan and a 2007 Chevrolet Silverado pickup) with real validated mass distribution, available in LS-DYNA keyword format and convertible to a simulation-ready collision mesh. This would replace the single-silhouette box proxy with genuinely distinct, citable vehicle geometry per class.
- **Physical-viability audit dashboard.** A direct instantiation of PVWM’s “physically viable” criterion: automated checks for mass conservation, momentum conservation, MPM Jacobian/volume-ratio bounds, zero-penetration contact, and energy monotonicity, applied to every run rather than asserted informally.
- **Failure-mode classification with violation magnitude.** An explicit failure-mode hierarchy (stuck, slide, topple, float) reported with the magnitude by which a run crosses each threshold, rather than a binary verdict.
- **Graph Network Simulator (GNS) surrogate.** A lightweight GNS trained on the completed simulation sweep as a fast surrogate for untested (depth, velocity, vehicle-class) combinations, connecting the L0/L1/L2 ladder to a learned, still physically-grounded fourth level.
- **A second independently-sourced real scene.** [FLAG: Verify exact citation and public availability of the FRED flooded-road dataset before including it as a committed future-work item rather than a candidate.]
- **DesignSafe-CI dataset publication.** Publish the validated real-scene dataset and simulation outputs with a DOI once the real-scene sweep is verified, rather than publishing the synthetic pilot data.

ACKNOWLEDGMENT

This work was supported by the National Science Foundation under NSF REU Site: Cyberinfrastructure Research for Societal Advancement (SCIPe-AI), Award #2447887, hosted at the Texas Advanced Computing Center (TACC), The University of Texas at Austin. The author thanks faculty mentor Dr. Krishna Kumar and research mentors Hassan Iqbal, Cheng-Hsi Hsiao, and Cristian Moran for their guidance throughout this project, and Rosalia Gomez for program coordination and support.

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